Post-Quantum Mathematics

Part 5 - Primitives & Cryptanalysis: Cryptanalysis of Candidate & Non-Ratified Schemes

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This part covers attacks on algorithms that are not yet standards. That distinction matters, because an attack on a candidate is how the process is supposed to work. A scheme that fails during evaluation has done its job by failing early.

The period covered here contains one outright casualty. HAWK, the only lattice-based candidate in NIST's additional-signatures round, was withdrawn in July 2026. The rest of this part covers pressure short of withdrawal: security margins narrowed, cost estimates lowered, attack techniques extended to parameter ranges they previously did not reach.

One family is under noticeably more pressure than the others. Multivariate schemes make up four of the nine remaining signature candidates, and Sections 2 and 4 show why NIST singled them out for mandatory parameter adjustments.

Section 1 covers attacks on code-based schemes and Section 2 attacks on multivariate schemes. Section 3 covers the isogeny result, Section 4 the withdrawal and NIST's response, and Section 5 the tool that keeps parameter estimates current.

1. Attacks on Code-Based Schemes

Both results here target Classic McEliece, the scheme with the longest unbroken record in this collection. Neither breaks it, and both narrow the margin it has relied on.

Recent Developments

A quasipolynomial-time distinguisher, running in $n^{O(\log n)}$, exists for Classic McEliece public keys in an asymptotic regime, and extends to heuristic key-recovery and decryption attacks [9]. A distinguisher tells an attacker that a key is a Classic McEliece key rather than random. That is weaker than recovering the key, but it is the step from which key recovery is often built.

Holdout key-recovery attacks against all five parameter sets have been costed at between $2^{126.77}$ and $2^{145.22}$ bit operations [8]. Those figures sit below the classical reference levels NIST set for the corresponding security categories.

Neither result is a practical attack, and the same two papers appear in Part 3 from the scheme's own perspective. Read together, they say the margin is thinner than it was in July 2026, on a scheme two European governments recommend and ISO has standardized.

2. Attacks on Multivariate Schemes

Multivariate signatures are the largest group in NIST's remaining candidate pool and the one under the most sustained attack. The line of work below has been extended three times in about a year.

Current State

UOV is a foundational multivariate scheme that other candidates build on, and it has a documented line of algebraic key-recovery attacks that keeps widening [2]. An original 2025 attack applied only when one parameter was below twice another. It was improved twice by different groups, and then extended in August 2026 to the remaining parameter range.

That extension has a concrete target. Applied to SNOVA at its lowest NIST security level, it reduces estimated security to 2^{103} gate operations [2]. That is below the level the parameter set was chosen to provide.

SNOVA is separately targeted by a wedge product attack, which extends prior algebraic attacks by exploiting the scheme's particular block structure [1]. This report could not retrieve that paper's own text due to a file-encoding problem, so the citation supports the attack's existence and general approach, not its parameter-specific impact.

A different kind of attack applies to UOV as implemented rather than as designed. A 2026 paper demonstrates a physical fault attack that recovers part of the secret linear map by inducing faults during operation [5]. It then transforms the public key system to complete the recovery. Fault attacks require physical access, which puts them in a different threat model from the algebraic work above.

3. The Isogeny Result

One new attack on the underlying isogeny problem, recorded here because its practical impact is nil and that is worth stating plainly.

Recent Developments

An unconditional attack solves the supersingular isogeny problem in time and memory $p^{2/5+o(1)}$, using a technique called bivariate multipoint evaluation [7]. It is of theoretical interest only and does not threaten practical isogeny cryptosystems such as SQIsign.

4. The HAWK Withdrawal and NIST's Response

The one candidate that did not survive. The manner of its withdrawal is as notable as the fact.

Recent Developments

HAWK was withdrawn on 29 July 2026 [6]. It was the sole lattice-based candidate advanced to NIST's third round of additional digital signatures. The attack reduced key recovery to a shortest-vector problem in roughly half the expected dimension, halving effective security margins for the proposed parameters.

The attack was AI-assisted, and the source attributes the assistance to Anthropic [6]. This is the first entry in this collection where an AI system contributed to breaking a cryptographic candidate rather than to analyzing one.

Its withdrawal also removes the only lattice-based scheme from the additional-signatures round, leaving that competition entirely to multivariate, isogeny and MPC-in-the-head constructions. Since the competition exists to diversify away from the lattice assumptions underlying ML-DSA, losing the lattice entrant arguably serves that goal.

RingSLIP, covered in Part 3, uses HAWK as a component.

Current State

NIST's round-3 requirements name the pressure directly. Submission teams had to address known cryptanalytic vulnerabilities by 14 August 2026 [3]. The requirement text specifically cites adjusting multivariate parameters to mitigate exterior product and small-field attacks. Naming one family in a general requirement confirms which class is under the most active pressure.

5. Keeping Parameter Estimates Current

A parameter set is only as good as the attack knowledge it was chosen against. This section covers the tool that tracks the difference.

Current State

The Lattice Estimator is the field's standard tool for estimating the security of parameter sets built on the Learning With Errors problem [4]. It is maintained specifically to track state-of-the-art attacks, so that parameter choices can be validated against current cryptanalysis rather than against whatever was known when they were chosen.

That maintenance is the point. Every scheme in Parts 1 and 3 that rests on lattice assumptions depends on someone continuing to do it.

About This Report

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Sources

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4. Lattice Estimator - https://lattice-estimator.readthedocs.io/en/latest/readme_link.html (undated)
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